

# Red-shift

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know.

|    | QUESTION                                                                                  | ANSWER                                                                                                | ✓                        |
|----|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------------------------|
| 1  | What is red-shift?                                                                        | The increase in the observed wavelength of light from galaxies that are moving away.                  | <input type="checkbox"/> |
| 2  | Explain why light from distant galaxies is red-shifted.                                   | The galaxies are moving away from us.                                                                 | <input type="checkbox"/> |
| 3  | Describe how the red-shift of a galaxy depends on its speed.                              | The faster it is moving away, the greater the red-shift.                                              | <input type="checkbox"/> |
| 4  | Describe how the speed of a galaxy depends on its distance from us.                       | The further away it is, the faster it is moving away.                                                 | <input type="checkbox"/> |
| 5  | Describe the Big Bang theory.                                                             | The Universe began from a very small, hot, dense region and has been expanding ever since.            | <input type="checkbox"/> |
| 6  | Explain why red-shift is evidence for the Big Bang theory.                                | It shows that distant galaxies are all moving away, so the Universe is expanding from a small region. | <input type="checkbox"/> |
| 7  | Name two things about the Universe that scientists still do not understand.               | Dark energy and dark matter                                                                           | <input type="checkbox"/> |
| 8  | State what observations of supernovae since 1998 suggest about the Universe.              | Its expansion is accelerating.                                                                        | <input type="checkbox"/> |
| 9  | State what happens to the observed wavelength of light from a galaxy moving away from us. | It increases.                                                                                         | <input type="checkbox"/> |
| 10 | State what the red-shift of distant galaxies provides evidence for.                       | That the Universe is expanding (the Big Bang theory).                                                 | <input type="checkbox"/> |